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***Cis*-Silicon Phthalocyanine Conformation Induces *J*-Aggregated Nanosphere with Unique Near-Infrared Absorbance and Fluorescence Enhancement: A Tumor Sensitive Phototheranostic Agent with Deep Tissue Penetrating**

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† Electronic supplementary information (ESI) available: Additional figures.

Abstract: Organic phototheranostic nanomedicines with optimized Near-infrared (NIR) biological transparent window (700–900 nm) are highly desirable for diagnosis and treatment of deep-seated tumors in clinic. As an excellent organic photosensitizer for photodynamic therapy (PDT) with outstanding photo- and thermo-stability, phthalocyanines (Pcs) have been used as building blocks of single-component nanomedicines. However, all the thus far reported Pc based single component self-assemblies are of the *H*-aggregate nature, to the best of our knowledge. This results in the simultaneously self-quenching of fluorescence emission and photodynamic activity as well as a greatly reduced tissue penetration due to the blue-shifted absorption. In the present work, intramolecular hydrogen bonding was formed between the two long and flexible axial NH₂-terminated diethylene glycol ligands of the amphiphilic SiPc molecule (SiPc-NH₂) in solution, leading to the employment of a *cis*-conformation of this molecule according to the ¹H-NMR spectroscopic result, which as building block then further self-assembles into the monodispersed nanospheres (SiPcNano) with a *J*-aggregation nature on the basis of electronic absorption spectroscopic result. As a consequence, SiPcNano exhibits significantly enhanced red-shifted absorption into the NIR range of 750~850 nm and fluorescence emission. This, in combination with the increased photodynamic effect for SiPcNano triggered by the protonation of amine groups due to the acidic nature of tumors, endows effective synergistic NIR photodynamic and photothermal effects in different cancer cells and thus effective inhibition of tumor growth in A549 bearing mice on the basis of a series of *in vitro* and *in vivo* evaluations. The present result provides a new approach for constructing novel single-component NIR organic nanomedicines for multifunctional cancer therapy.

Keywords: phthalocyanine, *J*-aggregation, self-assembly, Near-IR, phototheranostic Agent

Introduction

Nowadays, along with the increasing insight into the attractive features of nanomaterials for biomedicine applications, remarkable efforts have been dedicated to develop a variety of nanosystems for integrating different kinds of therapeutic modalities.¹⁻⁸ For example, the photo-therapeutic agents with optimized Near-infrared (NIR) biological transparent window (700–900 nm) permitting high tissue penetration are highly desirable and ideal for eradicating deep-seated tumors to avoid the intrinsic light absorption of tissues.⁹⁻¹⁵ Thus far, most efforts paid in this direction have focused on inorganic materials associated with their NIR light absorbance features and good photostabilities.¹⁶⁻²¹ However, the lack of good biocompatibility and long-term toxicity for these inorganic materials prohibits their translation into practical clinical applications. As a consequence, organic nanomaterials have received increasing attention as the promising alternatives due to their decent biocompatibility and dedicated synthesis features. In 1995, Chen and co-workers developed the indocyanine green (ICG) dye-based theranostic nanoplateforms, which showed good NIR-driven therapeutic properties.²²⁻²⁵ This was followed by the heptamethine dye-based theranostic nanoplateforms also with quite good NIR-driven therapeutic properties.²⁶⁻³⁰ Unfortunately, their practical clinical applications however was significantly retarded by their low photostability.^{10, 31-35} Obviously, developing organic phototheranostic nanomedicines with optimized NIR biological transparent window (700–900 nm) and good thermal, photo, and chemical stability is highly desired in this field.

Phthalocyanines (Pcs) possess an 18 π -electron conjugated planar architecture, exhibiting distinctive photophysical and photochemical properties in the red regions.³⁶⁻³⁹ This together with their outstanding photo- and thermo-stabilities as well as low dark toxicity and ease of chemical functionalization, renders a wide range of investigations over their photodynamic therapy (PDT)

and photothermal therapy (PTT) properties.⁴⁰⁻⁴⁴ Actually various methods have been developed for the construction of Pc-containing phototherapeutic nanomedicines with combined PDT and PTT properties. Integration of phthalocyanines and additional PTT agents into nanocarriers has been employed as one of the main approaches.^{12, 19, 45-49} This method, however, results in not only unsatisfactory biostability but also complicated toxicity because of the multiple components. As a result, self-assembly of phthalocyanine single component into organic nanomedicines has been utilized to develop corresponding phthalocyanine-based phototherapeutic nanomedicines in recent years. However, all the nanosystems composed of single Pc component seem to employ the *H*-aggregation nature, to the best of our knowledge. The face-to-face *H*-aggregate nature indeed favors the PTT properties of corresponding materials because the nonradiative relaxation pathway benefits the heat production,⁵⁰ which however is at the expense of the self-quenching of both the fluorescence and photodynamic activity associated with the blue-shifted absorption that is disadvantageous for tissue penetration. Fortunately, SiPc with two axial ligands on the opposite sides of the phthalocyanine plane might be able to present the molecules from the *H*-aggregation but to *J*-aggregation, paving a new way towards constructing novel phthalocyanine-based phototherapeutic nanomedicines with improved PDT and/or PTT functionalities.

Herein, amphiphilic SiPc with two long and flexible axial NH₂-terminated diethylene glycol ligands (SiPc-NH₂) was employed to fabricate phototherapeutic nano-materials. Formation of *cis*-conformation of the molecule in solution enable it as building block further self-assemblies into the monodispersed regular nanospheres (SiPcNano) with a diameter of ca.100 nm with a *J*-aggregation nature, [Figure 1](#), resulting in a red-shifted Q band even into the NIR range of 750~850 nm and unquenched fluorescence emission as well as photodynamic activity. This renders it possible to reach a synergistic PDT and PTT effects for this single-Pc based NIR theranostic

nanoplatfrom. Nevertheless, depending on the intrinsic pH-dependent behavior of the NH_2 moieties of SiPc-NH_2 , SiPcNano also possesses good acidic-natured tumor sensitive or targeting properties. As a total result, the effective synergistic anticancer PDT/PTT effects of SiPcNano was clearly disclose on the basis of systematic studies *via* a series of *in vitro* and *in vivo* evaluations.

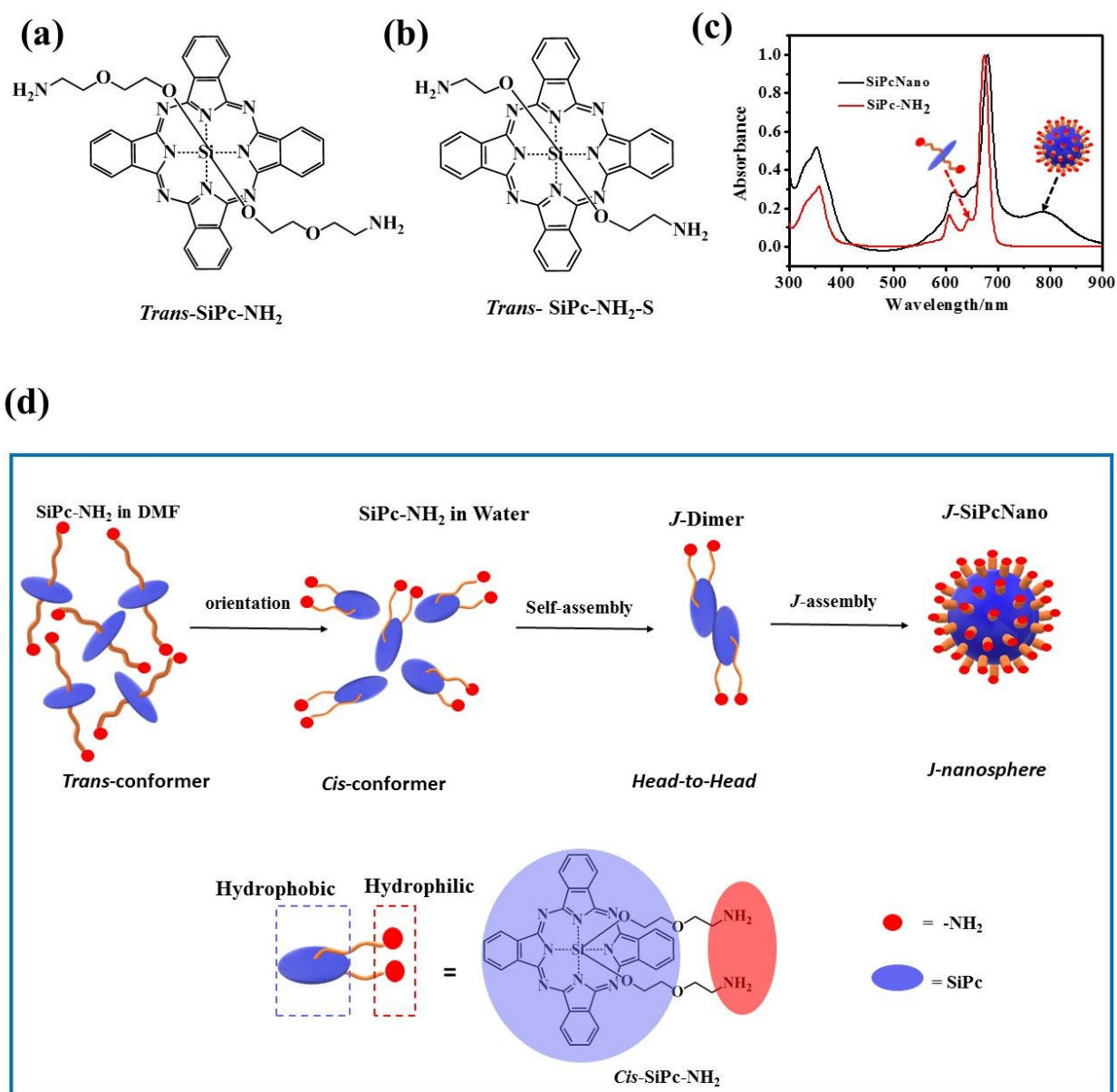


Figure 1. Structure of (a) SiPc-NH₂ and (b) SiPc-NH₂-S; (c) Absorption comparison of SiPc-NH₂ in DMF (red) and SiPcNano (black) in water. (d) Schematic illustration of self-assembly mechanism of SiPcNano formation in water.

Results and Discussion

Preparation of SiPcNano

Phototherapy, including photodynamic therapy (PDT) and photothermal therapy (PTT), is a kind of treatment that takes advantage of the incidence light to initiate intracellular photoactive agents to generate reactive oxygen species (ROS) or heat to induce apoptosis, which is of particular significance in the area of cancer treatment that is resistant to radio- or chemo-therapy. It has been realized the combination of PDT and PTT could enormously enhance the therapeutic efficacy in cancer treatment due to their synergistic effect, of which the oxygen independent mechanism of PTT enables it efficiently complement the decreasing efficiency of PDT caused by the hypoxia feature of tumors and oxygen over-consumption as the treatment progresses. Phthalocyanine (Pc) is one of the most promising PDT photosensitizers owing to its efficient singlet oxygen generation and excellent photostability. On the other hand, Pc-aggregate favors the nonradiative de-excitation pathway, which is heat-production and desirable for PTT. Despite of the abovementioned advantages, the simultaneously quenched fluorescence and PDT along with the blue-shifted absorption caused by *H*-aggregation of Pc is unfavorable for either imaging or treatment of deep-seated or large-sized tumors. Obviously, the development of NIR light-triggered drugs with combined PDT/PTT effect is urgently needed for clinical medicine.

Due to the effective π - π interaction between the neighboring molecules associated with their large macrocyclic π -conjugate system, most phthalocyanines with planar molecular structure tend

to form face-to-face nanosystems with *H*-aggregation nature.⁵¹⁻⁵² This induces the fluorescence quenching and blue shifted absorptions, thus reducing the tissue penetration ability. Fortunately, SiPc with two axial ligands on the opposite sides of the phthalocyanine plane might be able to prevent the molecules from forming the *H*-aggregate but to *J*-aggregate, paving a new way towards constructing novel phthalocyanine-based phototherapeutic nanomedicines with improved PDT and/or PTT functionalities. It is worth noting during the preparation of the present paper, a nano-system fabricated from SiPc compound under the regulation of surfactant was reported by Huang and co-worker, which did not exhibit PTT effect.⁵³ In addition, the lack of internal nano-structure prevents the clarification of structure-functionality relationship.

In the present work, SiPc-NH₂ with two long and flexible axial ligands terminated with NH₂ group was employed to fabricate nanomaterials for phototheranostic studies. In water or mixed H₂O-DMF, SiPc-NH₂ molecules self-assembly into the regular monodispersed nanospheres with a diameter of ca.100 nm (SiPcNano) *via cis*-conformer. In comparison with SiPc-NH₂ in DMF solution, the Q band of SiPcNano takes a significant red-shift even into the NIR range of 750~850 nm, revealing the *J*-aggregation nature of SiPcNano with the head-to-head molecular packing on the basis of Kasha model⁵⁴⁻⁵⁶

¹H NMR Evidence for *cis*-SiPc-NH₂

Due to the existence of *cis*- and *trans*-conformers of the SiPc molecules,⁵⁷ the molecular conformation of SiPc-NH₂ was studied by ¹H NMR spectroscopy towards clarifying the formation mechanism of the *J*-aggregated SiPcNano. As shown in the ¹H NMR spectrum of SiPc-NH₂ in CDCl₃, [Figure 2b](#), the signals due to the α - and β -protons of the Pc ring locating within the downfield region of $\delta = 8\sim 10$ ppm appear as AA'BB' splitting double peaks, giving two sets of

^1H NMR data. This is also true for the signals attributed to the protons of the axial ligands in the upfield region of $\delta = -2 \sim 2$ ppm. These results clearly reveal the existence of both *cis*- and *trans*-conformers of the SiPc molecules in solution with the group of signals locating in the upfield side to the *cis*-conformer and signals at the down-field side to the *trans*-conformer according to our previous findings.⁵⁷ At the first glance, it appears hard to understand the employment of *cis*-conformation for the SiPc molecule due to its relatively higher energy state in comparison with the *trans*-conformation. This, however, could be easily rationalized by the formation of intramolecular hydrogen bond between the two long and flexible axial $-\text{NH}_2$ groups as illustrated in [Figure 2a](#). To further testify this speculation, the ^1H NMR spectrum of SiPc- NH_2 was also recorded in CD_3OH . As displayed [Figure 2c](#), the group of up-field signals corresponding to the *cis*-SiPc- NH_2 get completely disappeared, while only the groups of down-field signals due to the *trans*-SiPc- NH_2 are observed in CD_3OD because of the formation of intermolecular hydrogen bonds between the axial $-\text{NH}_2$ group and the solvent CD_3OD rather than the intramolecular hydrogen bond at this case. Actually, the group of signals due to the *cis*-SiPc- NH_2 could not be observed in $\text{DMSO}-d_6$ due to the intense coordination ability of the solvent DMSO molecules, [Figure 2d](#), which also prevents the formation of the intramolecular hydrogen bond. Additional evidence for this point comes from the lack of signals due to the *cis*-conformer for another phthalocyaninato silicon compound SiPc- NH_2 -S with two shorter axial groups, [Figures S1 and Figure S2† \(ESI\)](#). Obviously these studies clearly demonstrate the existence of *cis*-SiPc- NH_2 in solution, which in turn induces the formation of SiPcNano with the *J*-aggregation nature as detailed below.

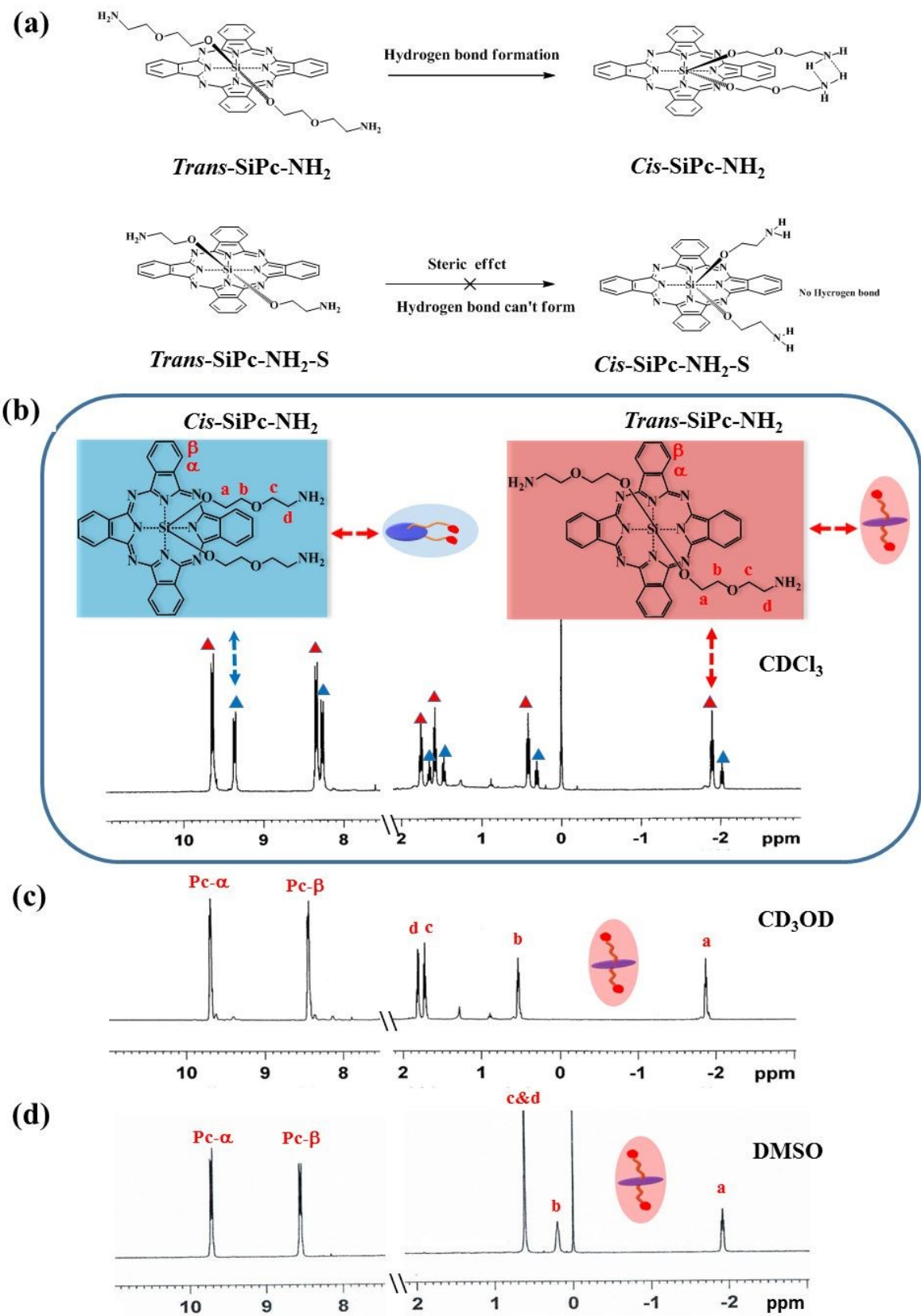


Figure 2. (a) Mechanism of intramolecular hydrogen bond formation of SiPc-NH₂ and SiPc-NH₂-S in aprotic solvent; Comparison of ¹H NMR spectra of SiPc-NH₂ in (b) CDCl₃, (c) CD₃OD and (d) DMSO-[d]₆.

Towards further clarifying the existence of both *trans*- and in particular *cis*-SiPc-NH₂ conformers in CDCl₃ solution, both temperature- and concentration-dependent ¹H NMR spectroscopic studies were carried out for SiPc-NH₂ in CDCl₃. As shown in Figure 3a, within the concentration range of 10⁻³ ~ 10⁻⁴ M, the group of upfield signals due to the *cis*-SiPc-NH₂ get gradually increased along with the decrease in the concentration, while those due to the *trans*-SiPc-NH₂ get decreased in an inverse manner, due to the increase in the intramolecular hydrogen bond number and decrease in the intermolecular hydrogen bond following the concentration change. This further verifies the close inter-relationship between the *cis*-SiPc-NH₂ and intramolecular bond formation. This seems to represent the first ¹H NMR evidence for the concentration regulated spatial conformation change of SiPc, to the best of our knowledge. Figure 3b shows the temperature dependent ¹H NMR spectra of SiPc-NH₂ in CDCl₃ recorded within the range of 298 ~ 313 K. Obviously, the groups of peaks attributed to *cis*-SiPc-NH₂ get gradually appeared along with the decreasing of temperature, indicating that the thermodynamically unstable *cis*-SiPc-NH₂ could be stabilized at low temperature.

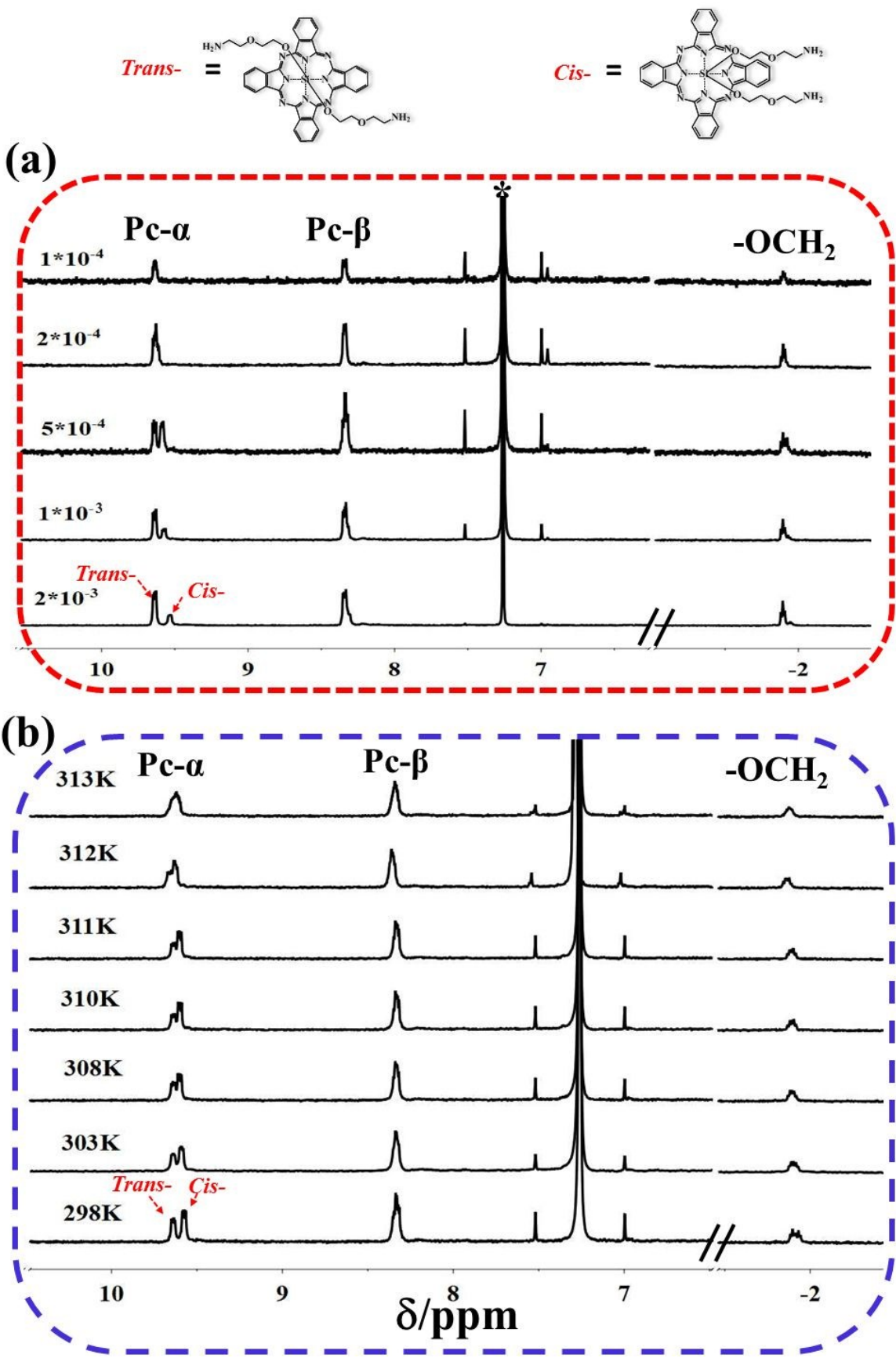


Figure 3. (a) ^1H NMR splitting patterns of SiPc-NH₂ in response to concentration change; (b) ^1H NMR splitting patterns of SiPc-NH₂ in response to temperature change.

***J*-aggregation nature of SiPcNano formed in Water**

Figure 1d illustrates the formation mechanism of SiPcNano with *J*-aggregate nature through the self-assembly of SiPc-NH₂ in water. At the first step, the two hydrophilic -NH₂ groups will point to the opposite side of the Pc moiety, leading to the employment of *cis*-conformation for the molecule, which acts as the hydrophilic part while the Pc macrocycle as the hydrophobic part on the other side for the *cis*-SiPc-NH₂ with enhanced amphiphilicity. Then at the second step, the adjacent monomer *cis*-SiPc-NH₂ form head-to-head aggregate with the hydrophilic part outstretching at the exterior to protect the interior hydrophobic part from exposing to the aqueous environment. Finally, the amphiphilic molecules of *cis*-SiPc-NH₂ rotate to arrange together to form the spherical nanoparticle with a hydrophilic shell and a hydrophobic core. This, however, is not the case for SiPc-NH₂-S due to its two short axial groups.

It seems strange to speculate that SiPc-NH₂ molecule in water could employ the *cis*-conformation at the first glance. However, as can be easily expected, in the case that water could dissolve the organic compound SiPc-NH₂, the formation of inter-molecular hydrogen bonds between the water molecule and the axial -NH₂ groups should render the SiPc-NH₂ molecule to take the *trans*-conformation. This, however, is not the case because of the intense hydrophilic nature of water molecule and the predominant hydrophobic nature of the SiPc-NH₂ molecule. Again as can be expected, when the SiPc-NH₂ compound was dispersed in water, the hydrophilic solvent molecules will surely get away from the hydrophobic part of the SiPc-NH₂ molecule (actually the phthalocyanine macrocycle π system) but closely interact with the hydrophilic -NH₂

groups in the axial ligands, enabling the two hydrophilic -NH_2 groups point to the opposite side of the hydrophobic Pc ring and rendering the employment of the *cis*-conformation of SiPc-NH₂ molecule in water. This in turn results in an enhanced amphiphilicity of SiPc-NH₂ molecule in water. For the purpose of protecting the hydrophobic phthalocyanine macrocycle from exposing to the water solvent molecules, the hydrophilic parts of *cis*-SiPc-NH₂ molecules would automatically arrange to the same direction/part to further closely interact with the water molecules in an aqueous environment, which in turn induces the formation of SiPcNano with *J*-aggregation nature. Figure 4e shows the concentration dependent absorption spectra of SiPcNano in water. As can be seen, SiPcNano exhibits the characteristic absorption bands of Pc with a Q band at $\lambda_{\text{max}} = 681 \text{ nm}$ and a Soret band at $\lambda_{\text{max}} = 354 \text{ nm}$, respectively. In addition, significant enhancement was observed for the absorption in the NIR region of 750-850 nm for SiPcNano, clearly revealing its *J*-aggregate nature and NIR photothermal heating potential. Furthermore, the aggregation behavior of SiPc-NH₂, as a function of water content, was studied in the mixed solvent system of H₂O-DMF, Figure 4f. In the mixed solvent with the H₂O volume fraction lower than 40%, the absorption spectra of SiPc-NH₂ display the typical molecular absorption characteristic bands. Upon increasing the H₂O volume fraction from 40 to 100%, aggregation occurs as disclosed by the remarkably decreased intensity for the Q band absorption accompanied by the simultaneous enhanced bathochromic peak at $\lambda_{\text{max}} = 790 \text{ nm}$, indicating the *J*-aggregation behavior. It is worth noting that under the same experimental conditions, *J*-aggregation does not occur for SiPc-NH₂-S with short axial ligands, confirming the significant role of the long and flexible axial ligands in SiPc-NH₂ in forming *J*-aggregated SiPcNano, Figure S6† (ESI).

In contrast to the *H*-aggregates that have nondetectable fluorescence quantum yield, fluorescence emission with Stoke's shift should be observed for *J*-aggregates according to

excitation theory.⁵⁸ As shown in Figure 4g, a 14 nm Stokes shift ($\lambda_{\text{max}} = 688$ nm) was observed in the fluorescence spectrum of SiPcNano, while that for the monomer SiPc-NH₂ amounts to only 2 nm ($\lambda_{\text{max}} = 676$ nm), giving additional support for the construction of SiPcNano with *J*-aggregate nature and in turn the satisfactory fluorescence emission for imaging probe. By contrast, the fluorescence emission was completely quenched for SiPc-NH₂-S in water, Figure S8† (ESI). The reason for this phenomenon is still not clear at this stage. However, it seems the *cis*-SiPc-NH₂ could stabilize the singlet excited state of SiPc to increase the fluorescence emission. Further study *via* molecular orbital calculation in near future is expected to clarify this problem.

The size and morphology of SiPcNano were revealed by Atomic force microscopy (AFM), scanning electron microscopy (SEM), and transmission electron microscopy (TEM) images reveal the uniform spherical nanoparticle morphology of SiPcNano with the diameter of ~100 nm, Figures 4a-4c. In addition, the size value of ~100 nm revealed by microscopic techniques is in consistence with the hydrodynamic diameter estimated by a dynamic laser light scattering (DLS), Figure 4d, which locates just in the optimal size distribution region (10 ~100 nm) for cancer therapy with high cellular uptake, enhanced permeability and retention (EPR) effect.

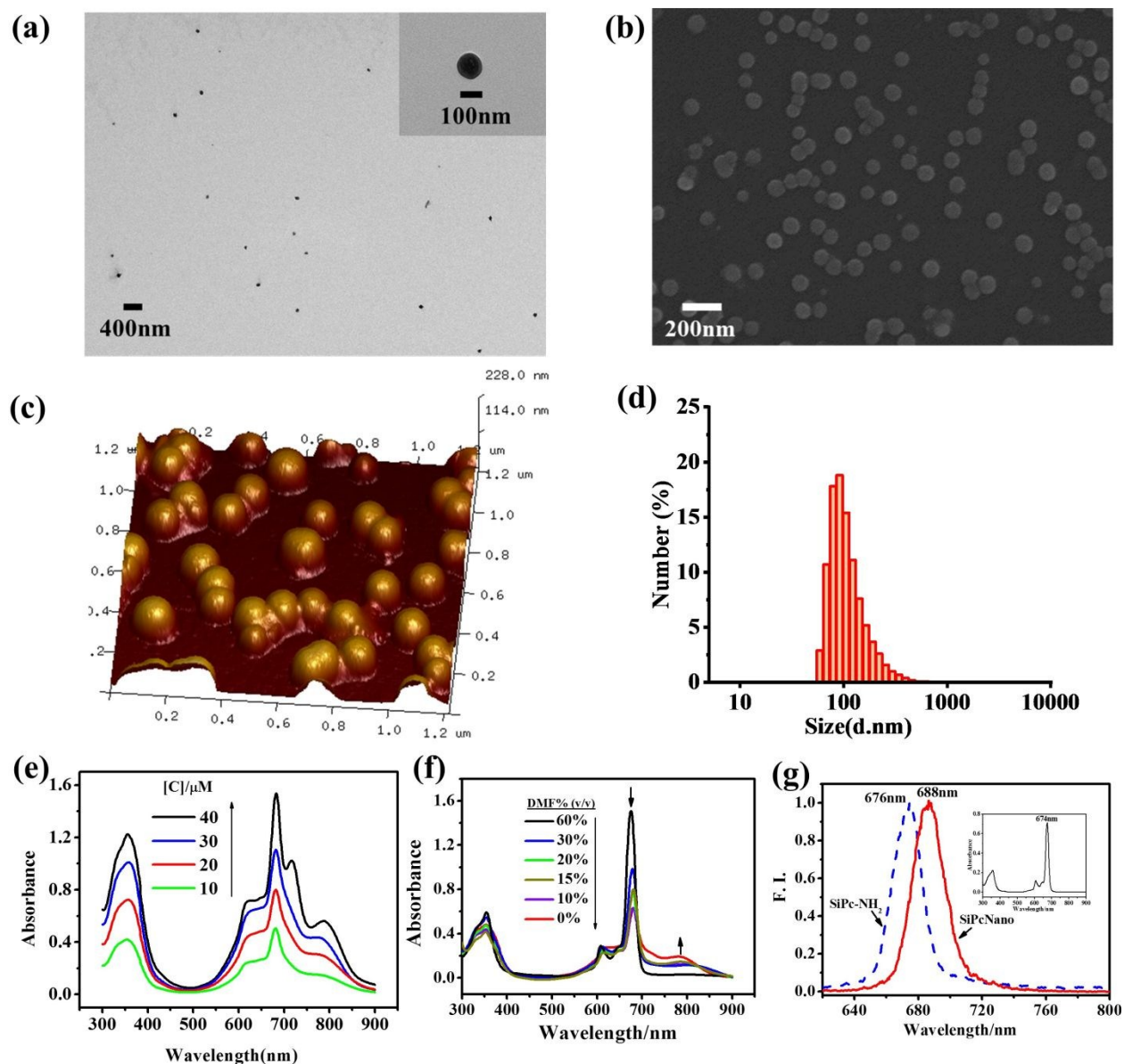


Figure 4. (a) TEM (b)SEM and (c) AFM photography of SiPcNano; (d) Size distribution of SiPcNano in water using DLS; (e) Absorption of SiPcNano at different concentrations in water. (f) Absorption spectra of SiPc-NH₂ in DMF-H₂O mixed solvent with different volume ratios. (g) Comparison of normalized fluorescence emission of SiPc-NH₂ in DMF (dash) and SiPcNano in H₂O (solid), inset shows SiPc-NH₂ absorption in DMF.

NIR Photothermal Properties of SiPcNano

The photothermal response of SiPcNano was recorded by a thermal imager. At first an 808 nm NIR laser was used for irradiation to certify its photothermal efficacy in NIR region. As shown in [Figure 5](#), the photothermal response of SiPcNano shows both concentration and laser power dependent characteristic. After 2 minutes of laser irradiation, the temperature of SiPcNano readily reached over 50 °C, actually the photoablation limit of living cells. Control experimental results under the same condition reveal the lack of obvious temperature change for pure water ($\Delta T < 3.5$ °C). By means of the procedure reported previously,⁵⁹⁻⁶⁰ the photothermal conversion efficiency of SiPcNano was measured and calculated to be 20.28%. Furthermore, as shown in [Figure 5g](#), no obvious reduction in the photothermal heating efficiency could be observed for SiPcNano after five cycles of irradiation with high power laser (808 nm, 14.5 W/cm²), demonstrating its satisfactory photothermal stability. Nevertheless, precipitation was not observed for SiPcNano in water after two weeks since preparation, indicating its long-term stability. These results disclose the high stability of SiPcNano as efficient NIR photothermal agent for PTT.

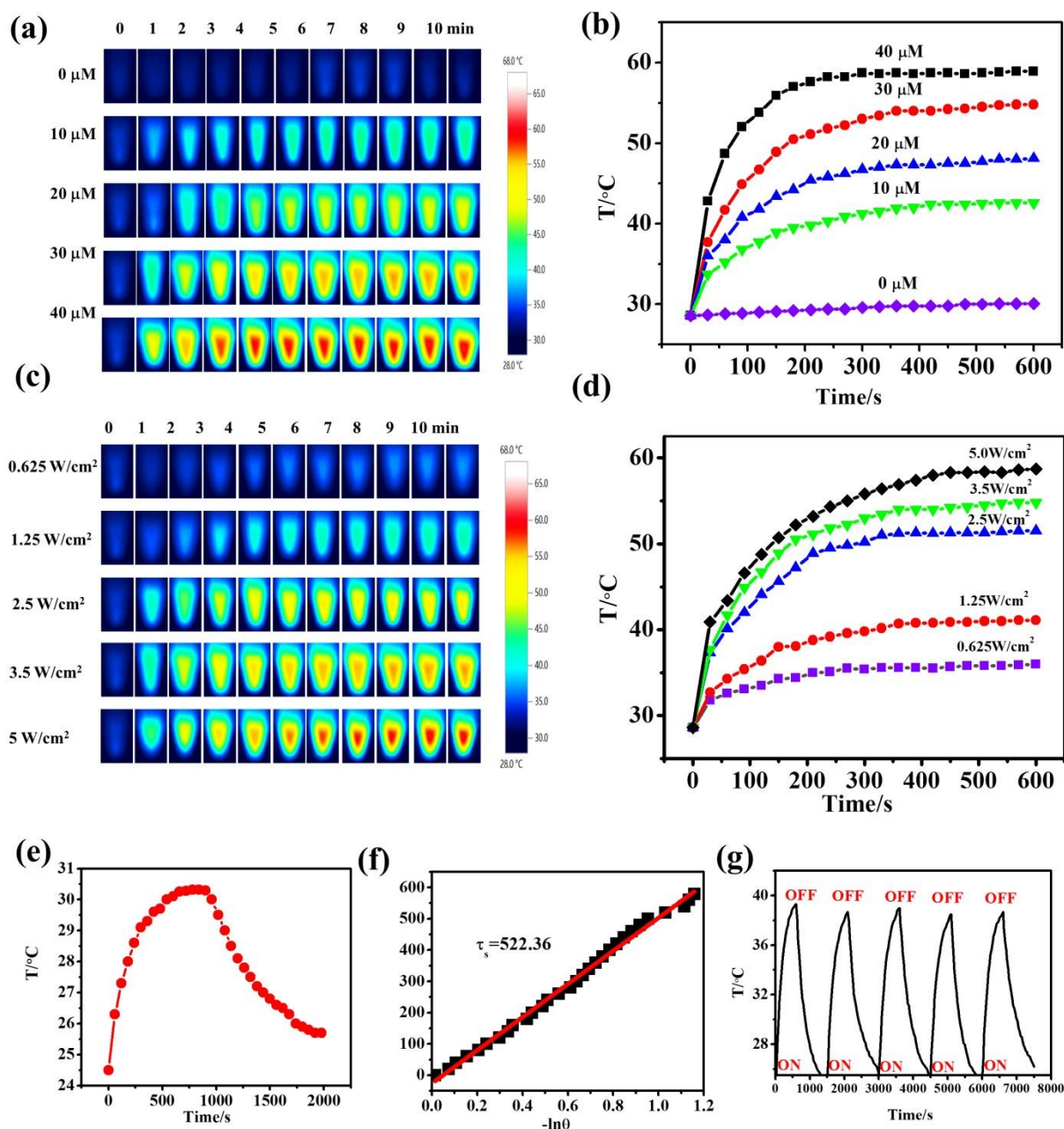


Figure 5. (a) Thermal images of SiPcNano at different concentrations under 808 nm laser irradiation ($3.5 \text{ W}/\text{cm}^2$) and (b) corresponding photothermal change curves. (c) Thermal images of SiPcNano ($30 \mu\text{M}$) under different 808 nm laser power and (d) corresponding photothermal change curves. (e) The temperature change of SiPcNano aqueous solution upon 808 nm laser irradiation and turn off at 15 min ($0.6 \text{ W}/\text{cm}^2$); (f) Plot of cooling time vs

the negative natural logarithm of the driving force temperature obtained from the cooling stage, as shown in Figure 5e; (g) The stability curves of SiPcNano in aqueous solution under 808 nm laser irradiation for 5 cycles.

Reactive Oxygen Species (ROS) Generation.

The steady-state method using 1,3-diphenylisobenzofuran (DPBF) as the scavenger was employed to evaluate the ROS generation capability of SiPcNano since the intensity of the absorption at 415 nm for DPBF would get gradually decreased along with the generation of ROS. To evaluate the NIR photosensitizing efficiency of SiPcNano, two sets of light source that consist of a halogen lamp and a color glass filter ($\lambda > 610$ nm or $\lambda = 750 \pm 10$ nm) were utilized for irradiation. Figure 6b compares the ROS generation efficiency of SiPcNano under different light sources in water. As can be seen, upon illumination by either red or NIR light source, ROS generation was clearly revealed. In addition, the consumption ratio of DPBF discloses the good capability of SiPcNano as NIR photosensitizer for PDT, Figure S9 and Figure S10 † (ESI). To the best of our knowledge, this seems to represent the first example of generating ROS by *J*-aggregated SiPc, indicating the photodynamic activity was not quenched within *J*-aggregated SiPc. This, however, is not the case for the *H*-aggregated Pc according to previous studies.⁵¹⁻⁵²

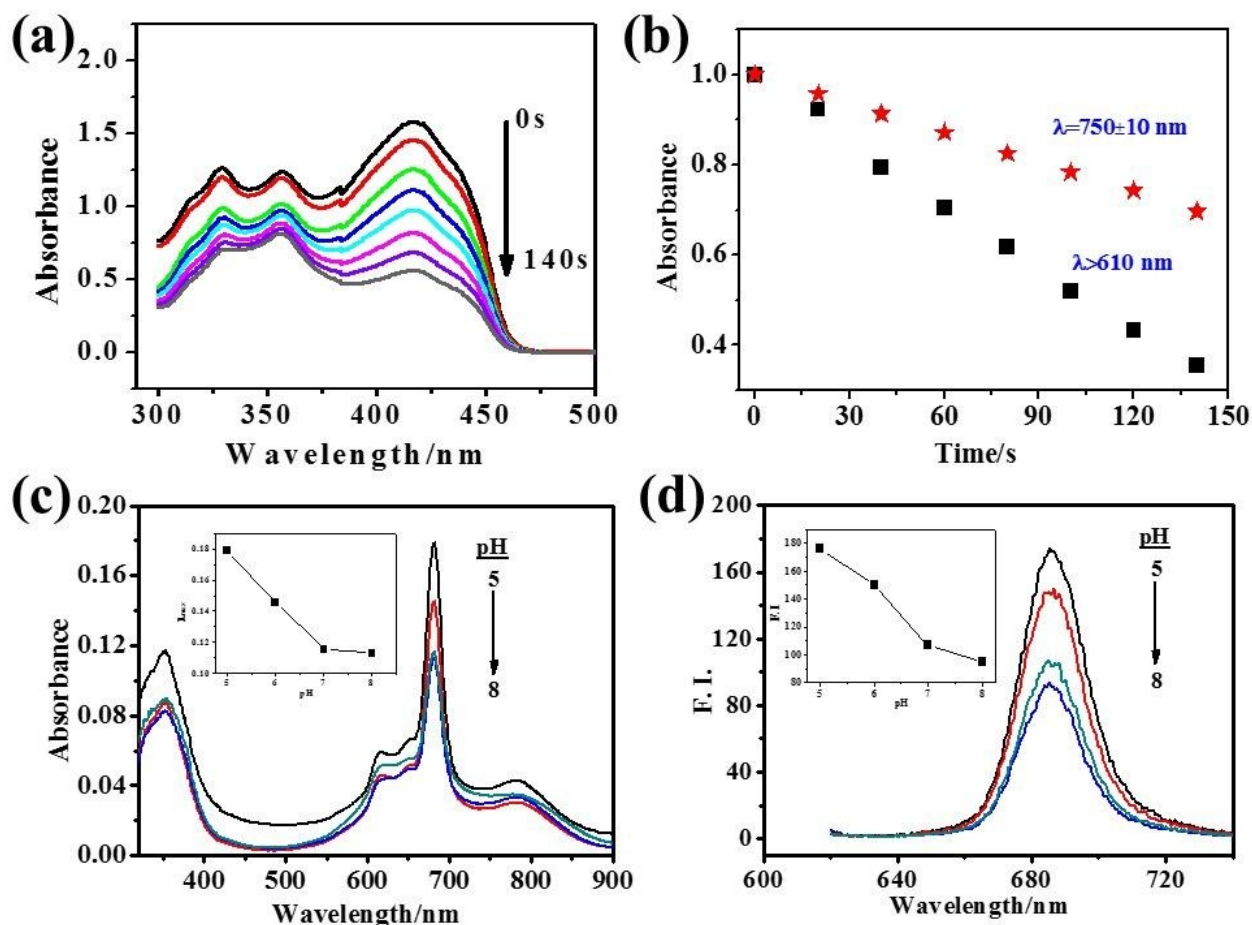


Figure 6. (a) Typical absorption spectra for the determination of singlet oxygen generation of SiPcNano in water exposed to light $\lambda > 610$ nm, using DPBF as scavenger. (b) Comparison of the photobleaching rate of DBPF exposed to light $\lambda > 610$ nm (square) and $\lambda = 750 \pm 10$ nm (star), respectively. (c) pH-Dependent absorption changes of SiPcNano; inset shows the Q band absorption changes. (d) pH-Dependent fluorescence emission changes of SiPcNano; inset shows the fluorescence $\lambda_{\max}$ changes.

pH-Dependent Properties of SiPcNano

It has been revealed that a relatively lower pH environment (acidic, pH 5~6) exists within tumors compared with the normal tissues (basic, pH 7.4) around them. Due to the acid-sensitive

moieties of the amine groups in SiPc-NH₂, different pH environment might change the stability and aggregation tendency of SiPcNano. As summarized in [Figures 6c-6d](#), SiPcNano exhibits a remarkable pH-dependent response in terms of either absorption or fluorescence emission. Both the absorption and fluorescence emission of SiPcNano in water get greatly increased upon reducing the pH value of the buffer solution from basic to acid, indicating partially release of SiPc-NH₂ from SiPcNano occurred due to the protonation of amine group under acid conditions, which will dramatically increase the PDT efficacy that is very sensitive to the concentration of SiPc-NH₂. These results suggest the great promise of SiPcNano as tumor sensitive drug with enhanced photodynamic activities for synergistic PDT and PTT.

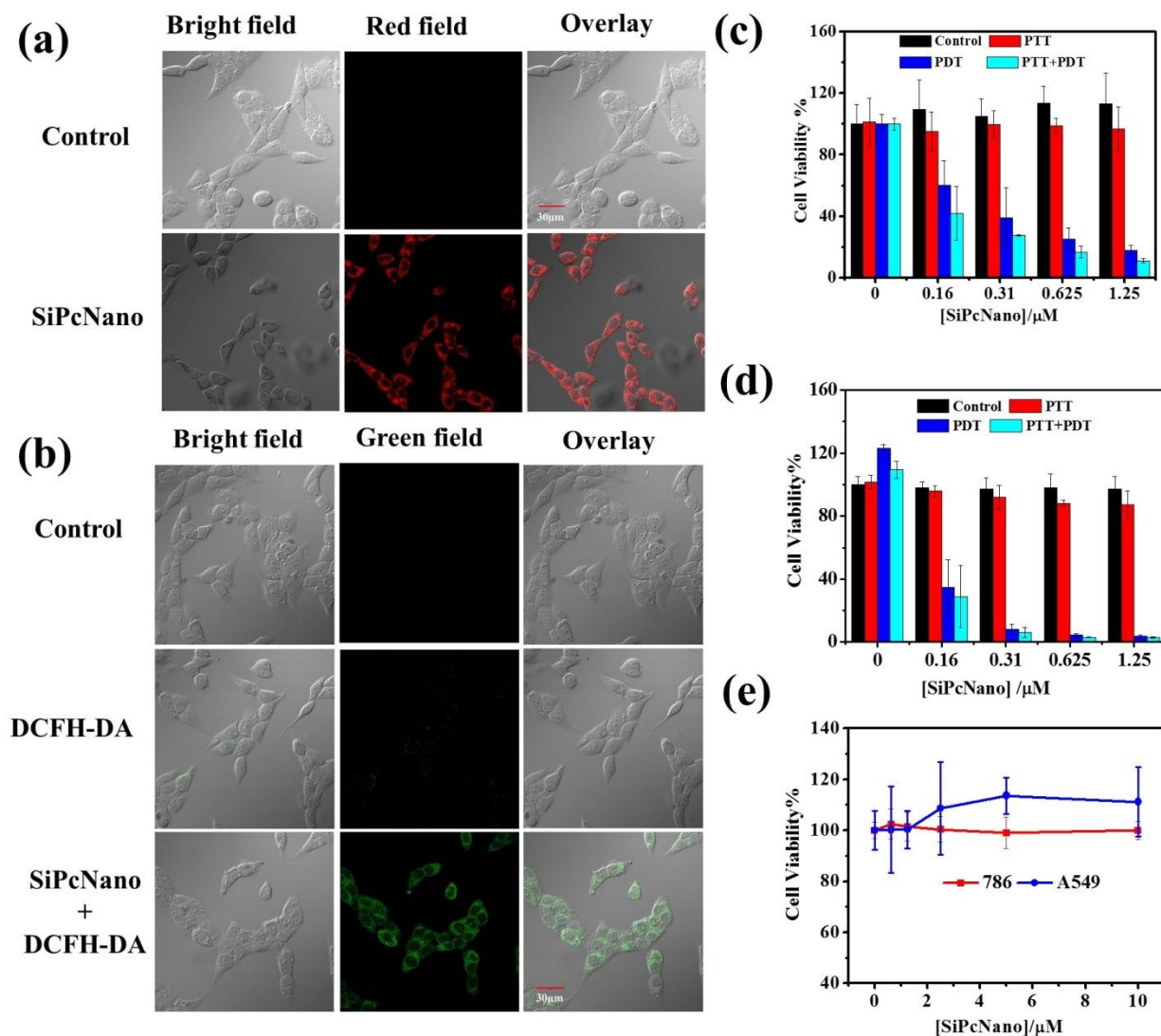


Figure 7. (a) Confocal fluorescence images of SiHa cells after being incubated by SiPcNano (10 μM) and PBS (control) for 4 h and (b) Confocal fluorescence images stained by DCFH-DA (10 μM) after being incubated by SiPcNano (10 μM) and PBS (control) for 4 h and irradiated by LED light ($\lambda \approx 660$ nm 0.2 mW/cm²) for 10 min. Cytotoxicity of (c) 786 and (d) A549 cancer cells treated with SiPcNano at different concentrations and irradiated by different light (808 nm laser at 3.5 W/cm² for PTT, LED light $\lambda \approx 660$ nm at 0.2 mW/cm² for PDT) for 5 min. (e) Dark toxicity of SiPcNano after being incubated for 24 h evaluated by CCK-8 assay.

***In Vitro* Anticancer Efficacy**

The *in vitro* anticancer effects of SiPcNano were studied in different cancer cell lines, namely SiHa, A549, and 768 cancer cells. First, both the intracellular fluorescence and intracellular ROS generation of SiPcNano were evaluated by a confocal microscope. As shown in Figure 7a, intracellular fluorescence images in cytoplasm were clearly observed after 4 h of incubation with SiPcNano at a concentration of 10 μ M. By contrast, negligible fluorescence could be observed for SiPc-NH₂ that containing Cremophor EL to inhibit aggregation, Figure S11† (ESI), which is consistent with the fluorescence emission observation in other amine substituted SiPc due to the amino moieties quench the singlet excited state of the phthalocyanines by photoinduced electron transfer (PET).⁶¹⁻⁶² It seems the *cis*-SiPc-NH₂ in SiPcNano could stabilize the singlet excited state of SiPc to increase the fluorescence emission. Further study *via* molecular orbital calculation in near future is expected to clarify this problem. Upon irradiating by a 660 nm LED light, obvious intracellular ROS generation was detected in cytoplasm using 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA) as the probe, Figure 7b. The intracellular localizations of fluorescence and ROS generation are matching well with each other. As a result, SiPcNano could not only enter cells but also give intracellular fluorescence and ROS for both cancer imaging and PDT.

Two light sources of NIR 808 nm laser or LED red light $\lambda \approx 660$ nm were used to evaluate the *in vitro* anticancer efficacy, and the standard cell counting kit 8 (CCK-8) assay was conducted to assess the cell viability. The cells were irradiated by an 808 nm laser, 660 nm LED light or both to evaluate the PTT, PDT or synergistic effect of PDT and PTT, respectively. We assume that the cell killing by an 808 nm NIR laser comes solely from the NIR photothermal heating since lights with wavelengths longer than 800 nm have insufficient energy to initiate a photodynamic

reaction,¹¹ while the LED light (0.2 mW/cm²) only gives PDT effect at such low watt density. Due to the synergistic therapeutic effect of PDT/PTT, the cell viabilities irradiated by a simultaneous application of both 660 nm red light and 808 nm lasers give enhanced cell killing effect, which was confirmed in both A549 lung cancer and 786 renal cancer cells. Furthermore, the dark toxicities of SiPcNano in both A549 and 786 cells exhibit satisfactory biocompatibility with the cell viabilities > 98% at the concentrations up to 10 μ M in the absence of irradiation, [Figure 7e](#).

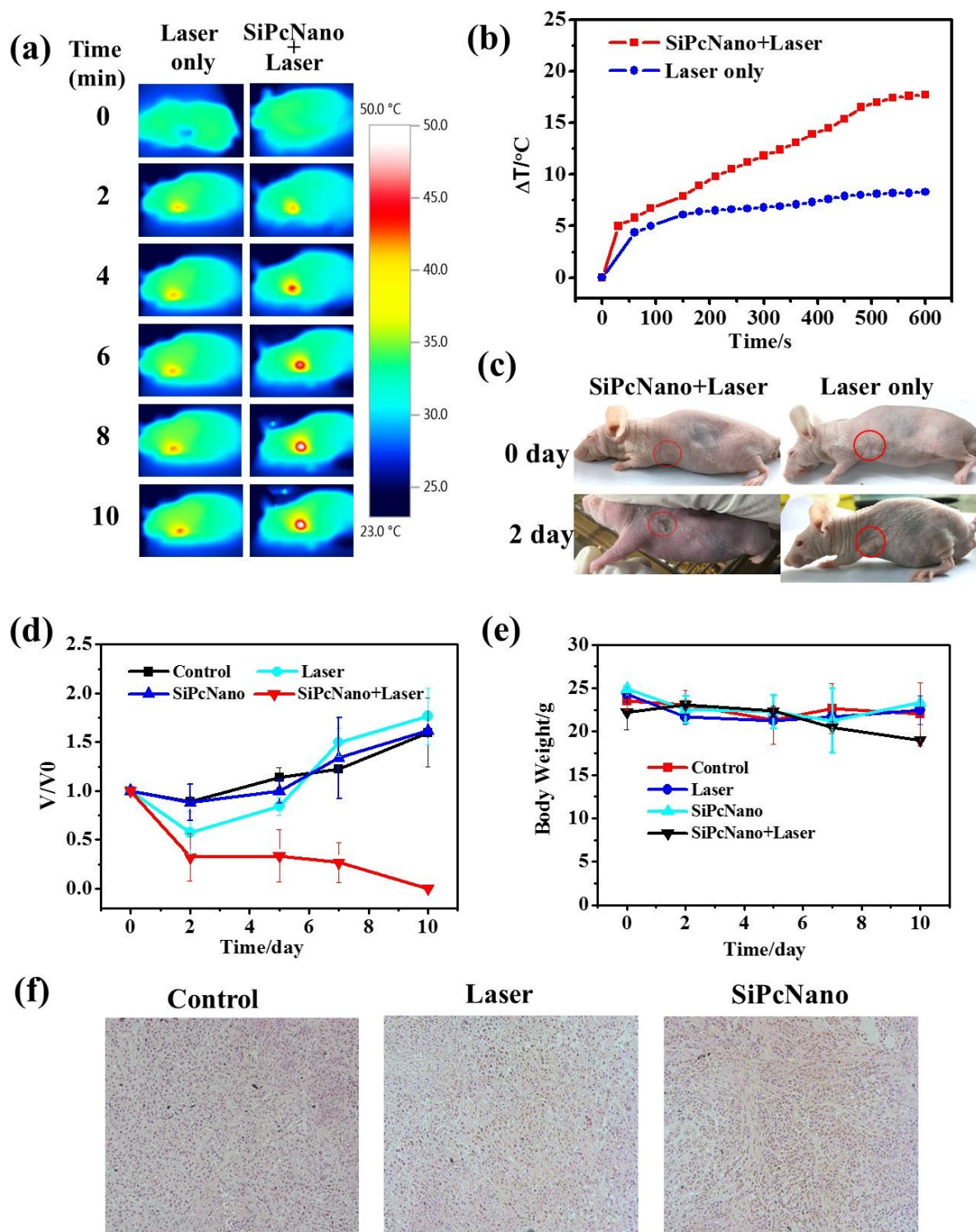


Figure 8. (a) IR photothermal images of A549 tumor-bearing mice exposed to 671 nm laser (2.5 W/cm²). (b) Tumor temperature change curves of mice along with irradiation time. (c) Typical photography of A549 tumor-

bearing mice before and after treatment. (d) Relative tumor volume curves of mice. (e) Body weight curves of A549 tumor-bearing mice in different groups. (f) H&E stained tumor tissues of different groups of mice at the 10th day post-injection.

***In Vivo* Anticancer Efficacy**

In vivo anticancer efficacy was investigated via intravenous injection of SiPcNano (1 mg/kg) into BALB/c nude mice bearing A549 tumors. For the synergistic PDT/PTT effect, the mice were irradiated with a 671 nm laser (2.5 W/cm², 10 min) and an infrared thermal imager was used to record the photothermal effect of tumor region every 1 minute. At wavelength of 671 nm, both PDT and PTT effect could be initiated synchronously under laser irradiation (2.5 W/cm²), [Figure S12† \(ESI\)](#). Therefore, 671 nm laser was chosen for synergistic PDT/PTT effect via a single laser irradiation. The infrared thermal mapping images demonstrate the temperature of the tumor rapidly increases over the photoablation threshold of 50 °C, causing irreversible destruction of cancer cells. This indicates that SiPcNano gets accumulated in tumor region and becomes responsible for the photothermal heating, [Figure 8a](#). In contrast, the control group that without using SiPcNano exhibits much lower temperature increase. The tumor growth rates and mice weights were monitored every 2-3 days. The results indicate that the irradiated tumor size of SiPcNano group gets greatly inhibited in comparison with the control and dark groups, verifying the excellent phototherapeutic effect of SiPcNano. As shown in [Figure 8c](#), the tumors became shrunken and purulent without any burn scar after only two days of post-irradiation, revealing the tumor therapy was caused by the intracellular damage of cancer cells due to the synergistic PDT/PTT effect of SiPcNano. At 10 days of post-irradiation, all the SiPcNano+laser treated tumors get completely disappeared, while those of the other groups kept growing gradually, [Figure 8d](#). At the same time, no obvious weight change was observed for all mice in different groups, indicating

the satisfactory biocompatibility of SiPcNano *in vivo*, Figure 8e. Furthermore, hematoxylin and eosin (H&E) staining were performed for the tumor tissues after 10 days of post-irradiation, showing no pathological difference between the control, laser only, and dark SiPcNano groups. This provides additional evidence for the good biocompatibility of SiPcNano in the absence of irradiation, Figure 8f. All these results clearly disclose the great promise of SiPcNano with *J*-aggregated nature for tumor treatment due to its enhanced NIR absorption for combined PDT/PTT, tumor sensitive property, and satisfactory biocompatibility.

Conclusion

Briefly summarizing above, homogeneous SiPcNano nanospheres with *J*-aggregate nature were constructed for the first time from amphiphilic silicon phthalocyanines with long and flexible axial ligands, exhibiting an increase in fluorescence and significant NIR absorption enhancement. The formation mechanism of the *J*-aggregated SiPcNano was clearly revealed by temperature-dependent and concentration-dependent ^1H NMR in addition to absorption and fluorescence spectroscopies. SiPcNano displays the NIR photothermal conversion efficiency of 20.28% as well as good photothermal stability upon irradiation by an 808 nm NIR laser. Both the *in vitro* and *in vivo* therapeutic results unveil the effective near infrared PDT/PTT synergistic effect for cancer treatment and fluorescence imaging properties of SiPcNano. The present result provides a new approach for constructing novel NIR organic nanomedicines with deep tissue penetration for multifunctional cancer therapy.

EXPERIMENTAL SECTION

Chemical and Materials. Polyoxyethylene castor oil (Cremophor EL, CEL) and 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA) were purchased from Sigma-Aldrich. Cell-

counting kit-8 (CCK-8) was obtained from DOJINDO LABORATORISE (Japan). All other reagents are in the purity of analytical grade and used without further purification.

Characterizations and Instruments. ^1H NMR spectra were recorded on a Bruker AVANCE 300 MHz spectrometer. Atomic force microscopy (AFM) images were obtained on a Dimension Icon (Veeco Instruments Inc.) and by noncontact optical microscope (VHX1000C, Keyence). Scanning electron microscopy (SEM) was observed using a HITACHI S-4800, resolution 1 nm (15 kV), vacuum in the sample chamber 10^{-4} Pa. Transmission electron microscopy (TEM) was observed using a HITACHI HT 7700. Ultraviolet–visible (UV–vis) spectra were measured on a Perkin Elmer Lambda-35 UV-vis spectrometer. Steady-state fluorescence spectra were obtained on a Hitachi F-2700 spectrofluorimeter equipped with a 450 W Xe lamp. Dynamic light scattering (DLS) was measured using a Zetasizer Nano ZS, Malvern Panalytical. The CCK-8 assay was performed on Infinite M200 Pro microplate reader.

Preparation of SiPcNano. SiPc-NH₂ was first dispersed in DMF to a concentration of 2 mM, and then slowly injected into deionized water with stirring using a syringe. SiPcNano aqueous solution was obtained after the injection was completed.

Photochemical and photophysical properties. The studies of $^1\text{O}_2$ generation in water were performed according to procedure described previously.^{42, 57} To detect the photothermal response of SiPcNano, 3 mL of SiPcNano at different concentrations was placed in a cuvette and irradiated by a 671/808 nm laser for 10 min. The temperature change was recorded every 30 s using an infrared thermal imager (Testo 872, Germany). The photothermal stability of SiPcNano was measured by irradiating the sample with an 808 nm laser for 10 min and turning off the laser for five cycles and recorded the temperature every 30 s.

Cell Culture. Human lung adenocarcinoma cells (A549), human kidney cancer cell (786), and human cervical cancer cells (SiHa) were used for *in vitro* biological evaluation. The cells were incubated in DMEM medium supplemented with 10% fetal bovine serum (FBS) and 1% penicillin and streptomycin (100 units mL⁻¹, streptomycin 100 µg mL⁻¹, respectively) at 37 °C under 5% CO₂.

Intracellular Fluorescence Imaging. Cells were seeded in 6-well plate and cultured overnight at 37°C under 5% CO₂. Then cells were incubated with SiPcNano (10 µM) in dark condition for 4 h. After incubation, the cells were rinsed with PBS three times. The fluorescence images were captured immediately with an Olympus FV 1200 confocal microscope, excited at 635 nm and monitored at 650–750 nm. The control group was incubated with only medium and acquired under the same conditions.

Detection of Intracellular ROS. The 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA) was used as an indicator for the detection of intracellular ROS. Cells were seeded on coverslips and cultured overnight at 37°C under 5% CO₂. Then, the cells were cultured with SiPcNano (10 µM) in DMEM for 4 h and the control group was cultured in pure DMEM. All cells were rinsed with PBS for three times after incubation, then 100 µL of DCFH-DA (10 µM) in PBS was added and cultured for another 15 min at 37°C under 5% CO₂. Then all cells were rinsed with PBS for three times again and exposed to light irradiation ($\lambda \approx 660$ nm, energy density = 0.2 mW/cm²) for 10 min. The fluorescence images were captured immediately with an Olympus FV 1200 confocal microscope, excited at 473 nm and monitored at 500-600 nm.

In Vitro Anticancer Efficacy. Cells were placed in 96-well plates (1 × 10⁴ per well) and cultured overnight at 37°C with 5% CO₂. The cells were then incubated with different concentrations of

SiPcNano for 24 h and the control group was cultured with pure medium. After incubation, the cells were rinsed with PBS three times and 100 μL of culture medium was added to each well before being illuminated. Then irradiation groups were exposed to LED red light $\lambda \approx 660 \text{ nm}$ (0.2 W/cm^2) or 808 nm (3.5 W/cm^2) laser for 5 min, dark groups were acquired under the same conditions without irradiation. The cell viability was assessed 24 h after irradiation using CCK-8 assay.

In Vivo Therapeutic Efficacy. All experiments using mice described herein were approved by the Animal Care and Utilization Committee of Shandong University and performed in accordance with applicable guidelines and regulations. The A549 cancer cells (approximately 10^7 cells per mouse) were subcutaneously injected into male BALB/c nude mice of 4–5 weeks old. When the tumor volumes reached approximately 50 mm^3 , the mice were randomized into four groups ($n = 3$) for control and evaluation of phototoxicity. The control and experiment groups were injected with PBS or SiPcNano (1 mg/kg, 100 μL) via the tail vein, respectively. At 24 h post injection, the mice treated with different therapy: Control (1): injection PBS + no irradiation; Laser (2): injection PBS + laser irradiation (671 nm, 2.5 W/cm^2 , 10 min); SiPcNano (3): injection SiPcNano + no laser irradiation; and SiPcNano + Laser (4): injection SiPcNano + laser irradiation (671 nm, 2.5 W/cm^2 , 10 min). The temperature change of tumors was detected using a thermal camera. After irradiation (day 0), the tumor size was measured every 2–3 days with a caliper and tumor volume was calculated according to formula: $\text{volume} = (\text{tumor length}) (\text{tumor width})^2 / 2$. The relative tumor volume calculated as V/V_0 , where V_0 is the initial tumor volume on day 0. The body weight of every mouse was measured by a digital balance.

At the end of the experiment, the mice were sacrificed and the tumors of the mice were retrieved. Then fixed in 4% formaldehyde solution at room temperature, dehydrated with ethanol, embedded with paraffin and sectioned into slices. After slicing and staining with H&E, observed their photos using an upright microscope (Nikon 90i).

Conflicts of interest

There are no conflicts to declare.

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Table of Content

Tumor sensitive *J*-aggregated self-assembly silicon phthalocyanine nanoparticle as synergistic anticancer organic phototheranostic nanomedicines with high tissue penetration.

